

FOC Getting Started Guide

Power Application Controller®





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1 INTRODUCTION

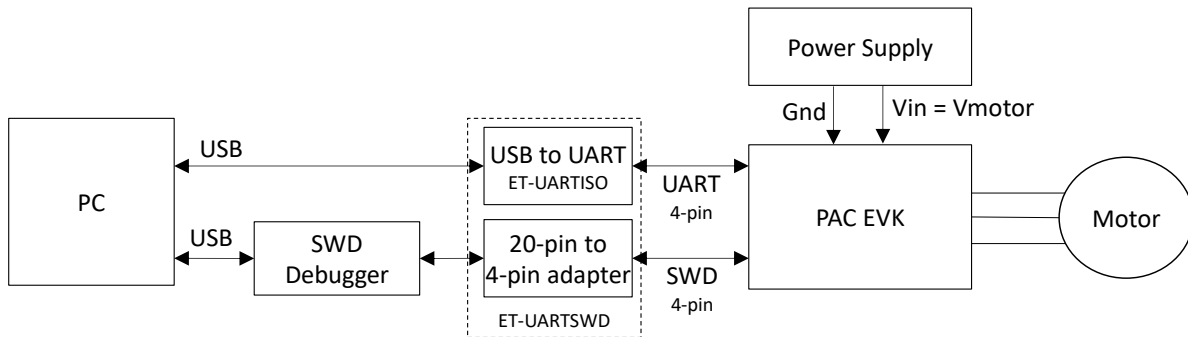
The PAC FOC Getting Started Guide is designed to get the PAC FOC FW up and running with a PAC EVK and motor. It assumes that the desired IDE has been installed and if using a PAC55xx, the appropriate PAC55xx IDE support files have been added. See the PAC FOC release notes for supported IDE versions.

2 DEVELOPMENT SETUP

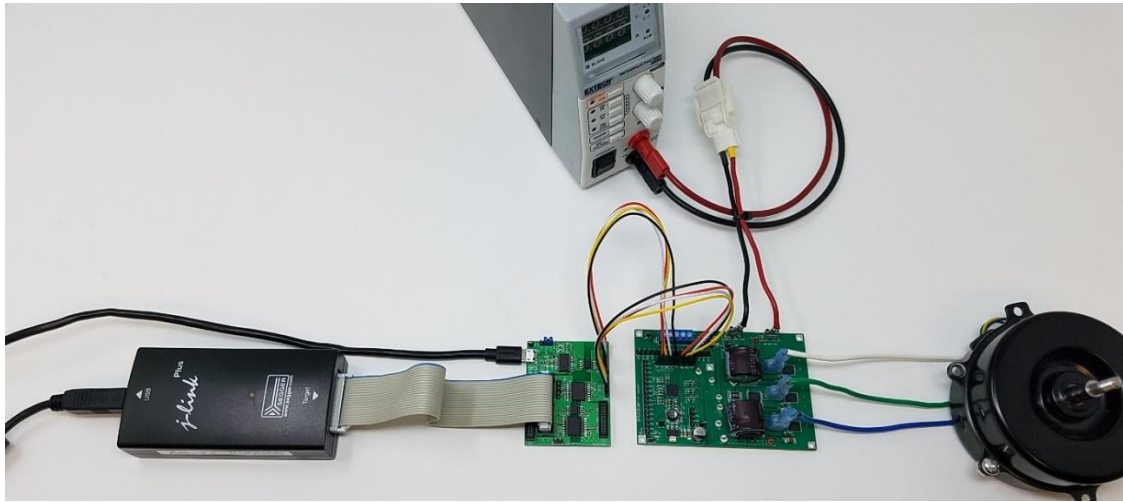
The first step is to setup the development environment to drive the desired BLDC/PMSM motor as shown in the figure below. The setup must include the following:

1. PAC EVK board
2. Power Supply
 - a. Should be capable of supplying V_{in} rated for the motor and the particular PAC EVK
 - b. Should be able to supply enough current as required by the motor
3. SWD debugger
4. Adapter Board(s) for connecting the SWD debugger and UART for FOC GUI communications
5. PC with desired IDE.

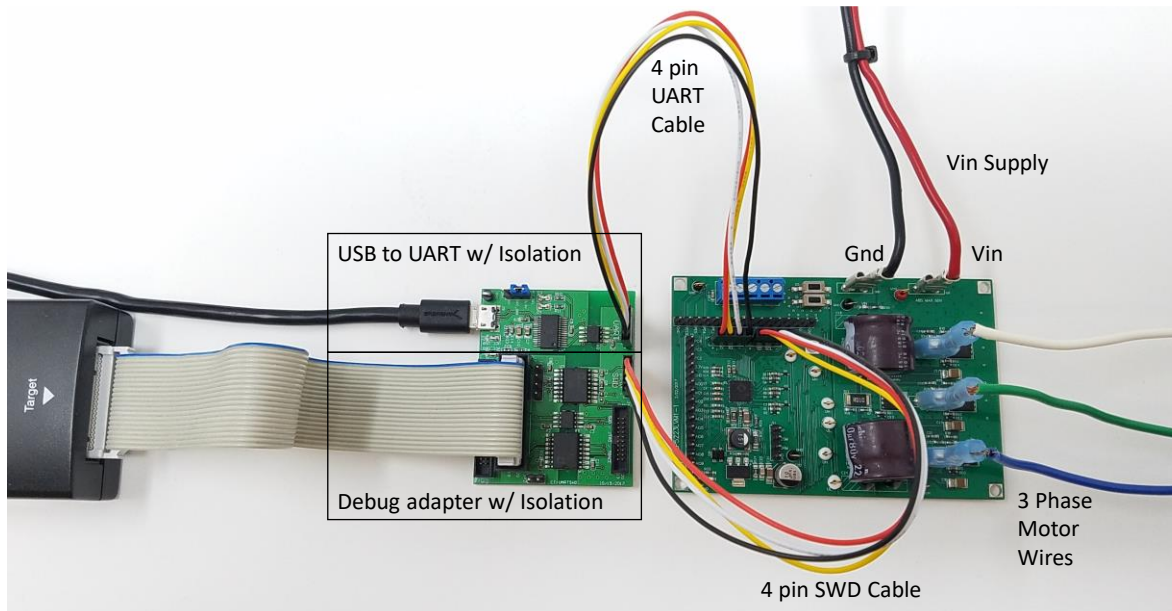
The figure below illustrates the different components.



The picture below illustrates the setup for a PAC5223 EVK using the ET-UARTSWD adapter board.



This picture is a close-up of the EVK connections for the PAC5223 EVK.



This getting started guide will walk through each connection step by step.

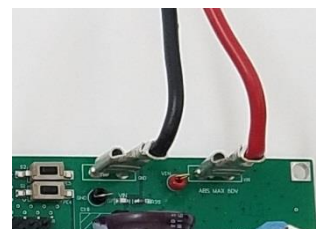
2.1 Motor Connection

Connect a 3 Phase BLDC/PMSM motor via spade connectors labeled U, V and W that are present on PAC EVKs. The image below shows the connection to the PAC5223 EVK.



2.2 Power Supply Connection

Connect a power supply to the Vin and Gnd input spade connectors on the PAC EVK according to the motor specification and the ratings of the PAC EVK. The image on the right shows the connection on a PAC5223 EVK.





Wait to turn on the supply until just before flashing the device with an IDE.

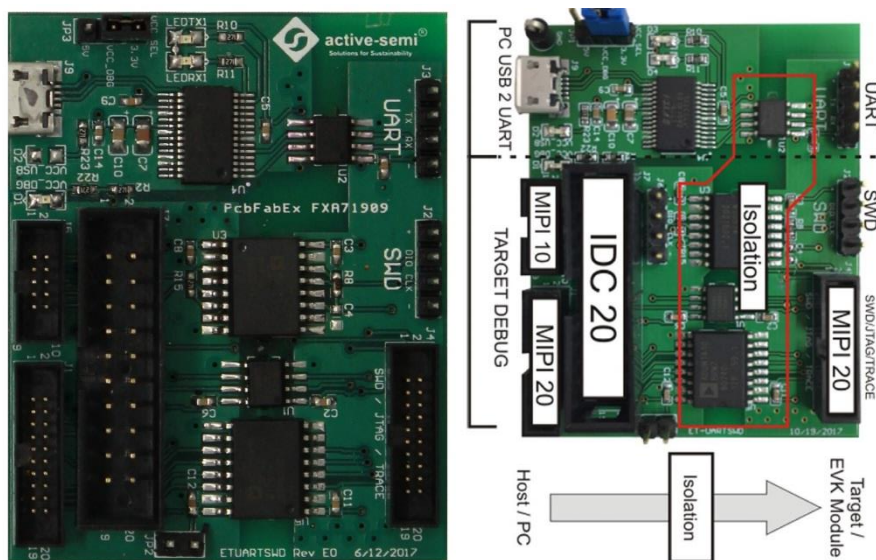
2.3 PC UART and Debugger Connection

To control the FOC FW, a Windows based GUI has been created that requires a UART connection to the PAC device. And, a debugger is required to flash the FOC FW to the PAC device. Two primary options are available for the UART and debugger connectivity:

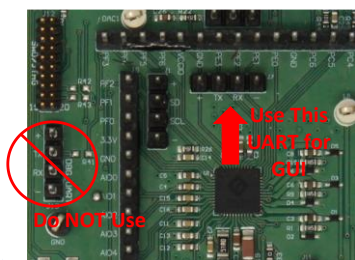
- 1) ET-UARTSWD combination adapter board
- 2) ET-UARTISO-1 with either ET-IARISO-1 or jumper SWD debugger connection

2.3.1 Option 1: ET-UARTSWD

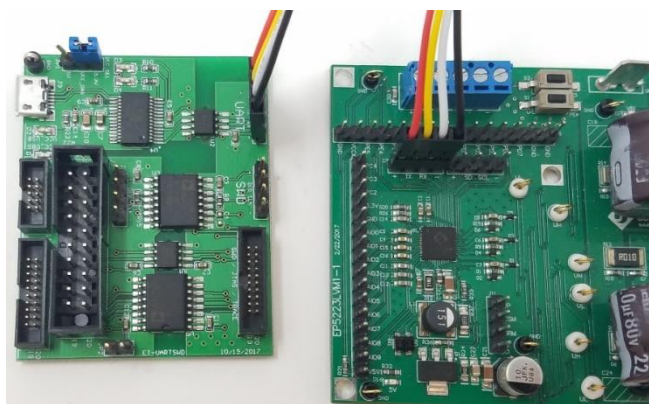
Using the ET-UARTSWD adapter board is the best method for UART and Debugger connectivity. New PAC EVKs are now shipping with the ET-UARTSWD combination adapter board that includes isolation for high voltage applications (see image below on left). The ET-UARTSWD has both a USB to UART function and connectors that adapt various debugger connector types to the 4-pin SWD header found on all PAC EVKs. It also includes an optional target side 20-pin MIPI connector for PAC55XX EVKs. Various aspects of the board are highlighted in the image below on the right.



To connect the UART, use a 4 wire cable to connect the PAC EVK 4-pin UART header labelled [+ TX RX -] to the ET-UARTSWD 4-pin UART header labelled [+ TX RX -]. The image on the right shows the connections for the PAC5223 EVK.

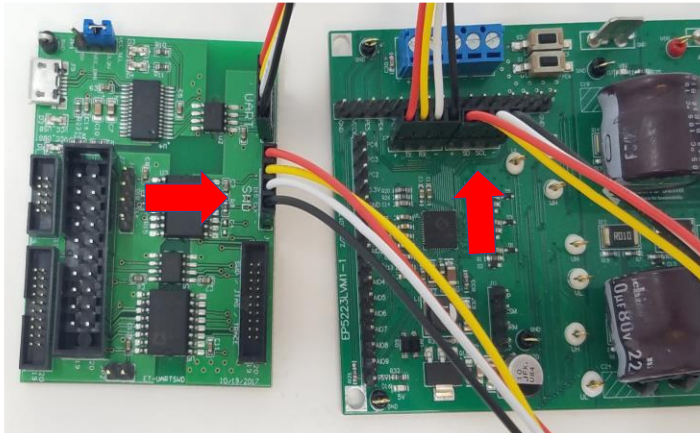


Note: On PAC55xx EVKs make sure the cable is not connected to the DBG_UART. The image shown on the left highlights the 2 UART headers on the PAC5523 EVK. The

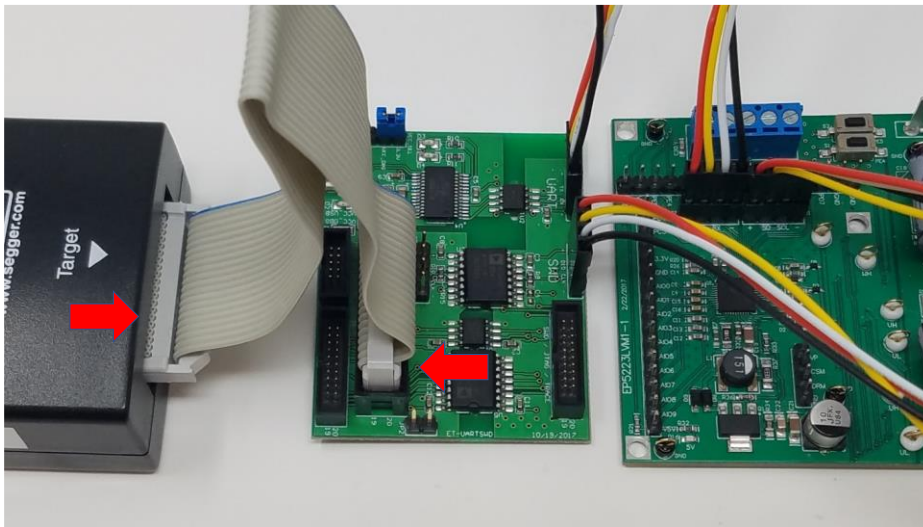


DBG_UART is primarily used for printf() via UART to a PC terminal applications.

Next, use a 4 wire cable to connect the PAC EVK 4-pin SWD header labelled [+ SD SCL -] to the ET-UARTSWD 4-pin UART header labelled [+ DIO CLK -]. The image below shows the connection for the PAC5223 EVK.



Now connect the 20-pin IDC ribbon cable of the SWD debugger to the appropriate PC/Host side connector on the ET-UARTSWD board. The image below shows a J-Link 20 pin IDC ribbon cable connection.

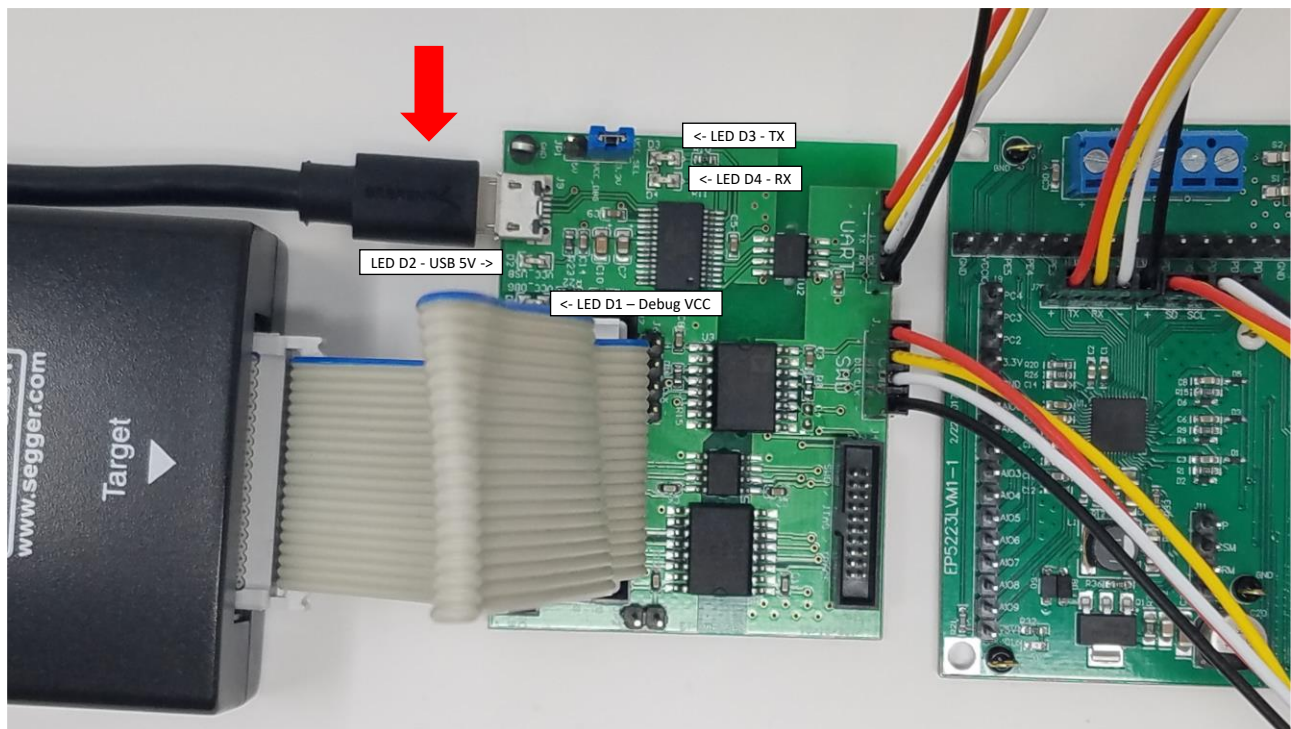


Notes:

- 1) The majority of debuggers sense the target hardware Vcc on pin 1 and can handle a 3.3v input. So, the most typical configuration for the ET-UARTSWD is where the power for the PC/Host side of the isolation is supplied by the USB to UART circuitry. For this configuration, the ET-UARTSWD jumper JP3 should be connecting 3.3v to VCC_DBG, which is the default position. If the debugger can't support this because
 - a. it supplies a voltage on pin 1,

- b. it doesn't support 3.3v input, or
 - c. if a USB connection to the PC is not being used, then consult the ET-UARTSWD User's Guide for powering the debugger Vcc.
- 2) Debuggers that have built in isolation are not supported, because the 2 isolation circuits will interfere with each other.
 - a. The ET-COLINK-1 debugger for CooCox includes isolation and so is not supported with ET-UARTSWD. ET-COLINK-1 can be directly connected to the 4-pin SWD header.
- 3) For debuggers with built in isolation, connect the 4 pins [Vcc, SD, SC, Gnd] directly to the SWD 4 pin header on the PAC EVK using jumpers.

Next connect a USB cable from the PC to the ET-UARTSWD adapter board as illustrated in the image below. If USB power is present, then LED D2 should light up. LED D1 should also light up indicating the debug Vcc is active. D3 and D4 LEDs will turn on and off based on UART TX and RX activity respectfully.

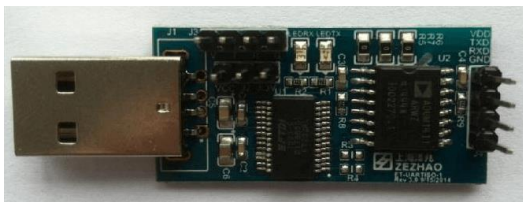


Finally, plug the debugger's USB cable into the PC.

Now proceed to paragraph 3 to build and flash the FOC firmware image.

2.3.2 Option 2: ET-UARTISO-1 with ET-IARISO-1 or jumper SWD connection

Another option for UART connectivity is to use the ET-UARTISO-1 adapter for USB to UART communications. In this case, SWD connectivity to the PAC EVK is handled separately. The ET-UARTISO-1 is shown below.

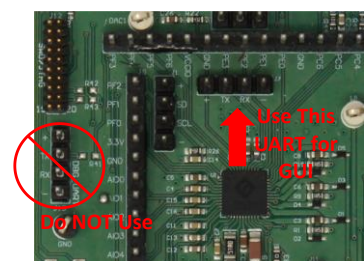


To connect the UART, use a 4 wire cable to connect the PAC EVK 4-pin UART header labelled [+ TX RX -] to the ET-UARTISO-1 4-pin UART header labelled [V T R G]. Note that V T R G stands for Vcc, Tx, Rx, and Gnd respectively. Then, connect the USB port to a PC.

Note:

- 1) The ET-UARTISO-1 can be plugged directly into a USB port on the PC, or used with a USB extension cable.

The image on the right shows the connection of the ET-UARTISO-1 to the PAC5223 EVK using a USB extension cable.



Note: On PAC55xx EVKs make sure the cable is not connected to the DBG_UART. The image shown on the left highlights the 2 UART headers on the PAC5223 EVK. The DBG_UART is primarily used for printf() via UART to a PC terminal applications.

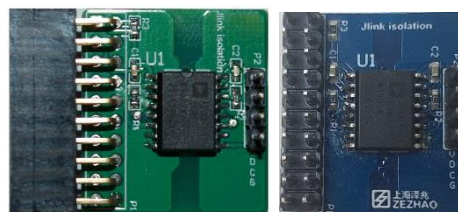
LED TX and LED RX LEDs will turn on and off based on UART TX and RX activity respectfully.

When using the ET-UARTISO-1, there are 2 options for SWD connectivity to the PAC EVK:

- 1) Use the ET-UARTISO-1 SWD adapter board
- 2) Connect the SWD debugger to the PAC EVK using jumper wires

2.3.2.1 ET-IARISO-1

For SWD debugger connectivity, the ET-IARISO-1 isolated SWD adapter board can be used to adapt the 4 pin SWD connector of PAC EVKs to most SWD debuggers that have a standard ARM 20-pin IDC connector. Two versions of the ET-IARISO-1 adapter board are shown on the right. The green one has a right angle connector, which can be directly plugged into the SWD debugger without a cable. The blue one has a



straight header, which the debugger 20-pin ribbon cable can be plugged into.

Because the debugger side of the ET-IARISO-1 requires power, either

- 1) The SWD debugger must be cable of supplying 3.3V power on Pin 1 of the 20-pin IDC connector.
- 2) Or, an external 3.3v supply must be connected to Pin 1 of the 20-pin IDC connector

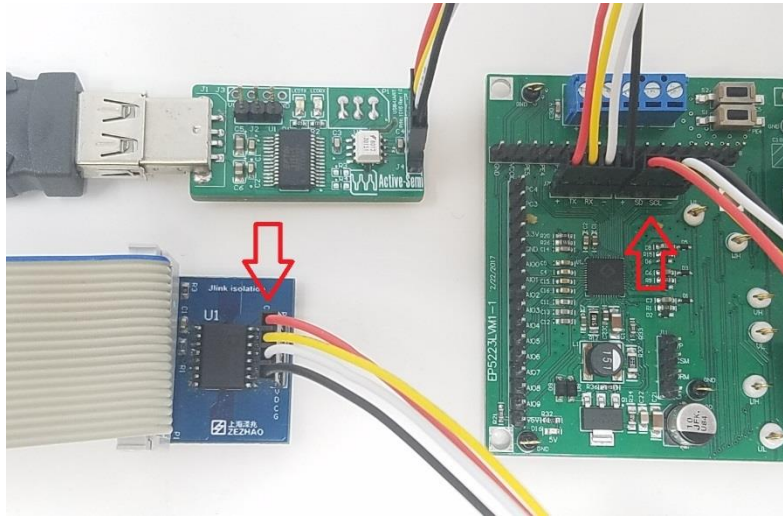
If these two options aren't possible, then consider the SWD debugger jumper method in paragraph 2.3.2.2.



Note:

- 1) Debuggers that have built in isolation are not supported, because the 2 isolation circuits will interfere with each other.
- 2) Note that Pin 1 on the ET-IARISO-1 has a square solder connection on the bottom of the adapter board.

Now, use a 4 wire cable to connect the PAC EVK 4-pin SWD header labelled [+ SD SCL -] to the ET-UARTISO 4-pin UART header labelled [V D C G]. V D C G stands for Vcc, SD, SCL, Gnd respectively. The image below shows the connection for the PAC5223 EVK.



Now connect the 20-pin ribbon cable of the SWD debugger to the ET-IARISO-1 board. Pin 1 of the ET-IARISO board is at the top and represented by the blue wire of the ribbon cable in the figure above. Connect the other end of the 20-pin ribbon cable to the 20 pin IDC connector on the SWD debugger.

Now proceed to paragraph 3 to build and flash the FOC firmware image.

2.3.2.2 SWD Debugger Connection Using Jumper Wires

Use of jumper wires to connect the SWD Debugger should be considered if any of the following are true:

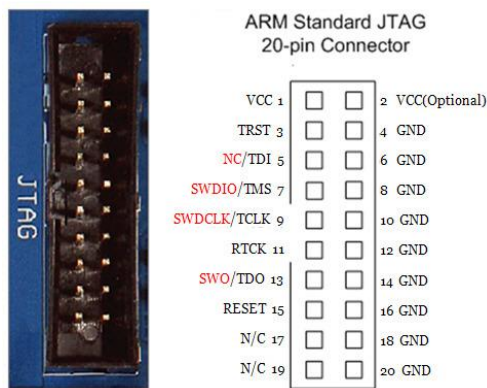
- 1) an ET-UARTSWD adapter is not available
- 2) an ET-UARTISO-1 is not available
- 3) an ET-UARTISO-1 is available, but 3.3V can't be supplied to pin 1 of the adapter
- 4) The SWD Debugger has built in isolation



IMPORTANT Notes:

- 1) When connecting a non isolated SWD Debugger to a high voltage PAC EVK using jumpers, care must be taken or injury and PC damage could occur. For PAC52xx EVKs, the default SWD signal voltage is 5V.
- 2) If the SWD debugger doesn't support 5V operation, do NOT connect the debugger via this jumper method.

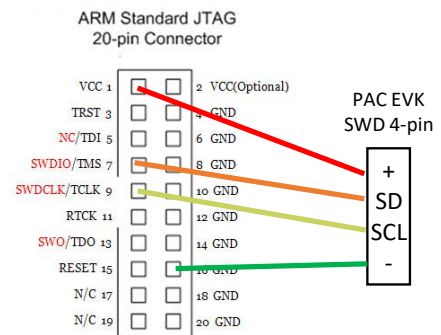
Connection of the SWD Debugger using jumper wires involves connecting 4 jumpers to the PAC EVK. A standard ARM JTAG/SWD 20-pin IDC connector pin out is shown below.



Connect VCC, SWDIO, SWCLK, and a GND using jumper wires to the 4 signals of the PAC EVK 4-pin SWD header labelled [+ SD SCL -]. This is illustrated in the block diagram on the right.

The picture below shows the actual connection of a J-Link debugger and the PAC5223 EVK. The jumper wires in the picture are as follows:

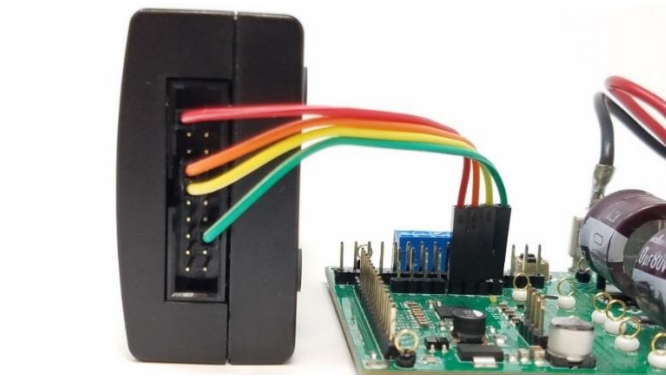
- Pin 1 - Vcc = Red
- Pin 7 - SWDIO = Orange
- Pin 9 - SWCLK = Yellow
- Pin 15 - Gnd = Green





FOC Getting Started Guide

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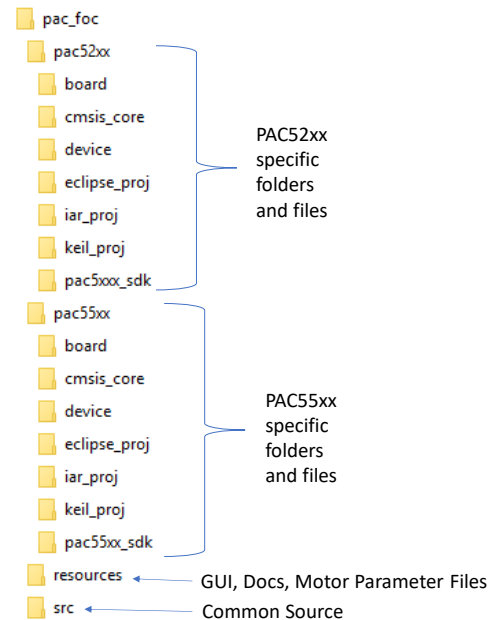
3 BUILD AND FLASH THE FIRMWARE

The next few steps will configure, build, and flash the firmware to the device on a PAC EVK.

3.1 Choose Device Family and IDE Project

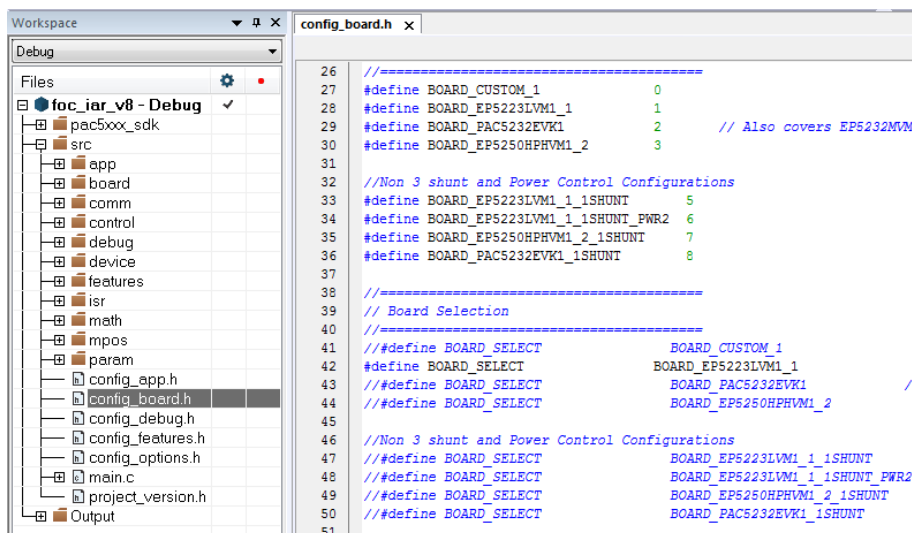
The PAC FOC FW contains common source used for various PAC devices and several IDEs; this code is stored in the src folder. The device dependent code and IDE projects are stored inside the device family folders. The figure to the right shows the folder hierarchy of the PAC FOC FW. First choose the device family, either PAC52xx or PAC55xx. Then, navigate down to the project folder for one of the IDEs: Keil, IAR, or Eclipse. Using the desired IDE, open the project contained in the associated folder.

NOTE: If using IAR v7.x, it must be updated with new configuration files. See `.resources\ide_support\iar` for instructions.



3.2 Configure Firmware

Now, navigate within the folder view of the IDE project to the src folder as shown in the figure below. Several configuration header files are provided to configure the FOC FW. Open the `config_board.h` file and navigate to the “Board Selection” portion of the code. Uncomment the `BOARD_SELECT` for the desired PAC EVK, and comment out all other `BOARD_SELECT` lines. In the example below, the EP5223LVM1-1 (aka PAC5223 EVK) has been chosen.





3.3 Build and Flash the Firmware

To build and flash the firmware image, follow the steps below:

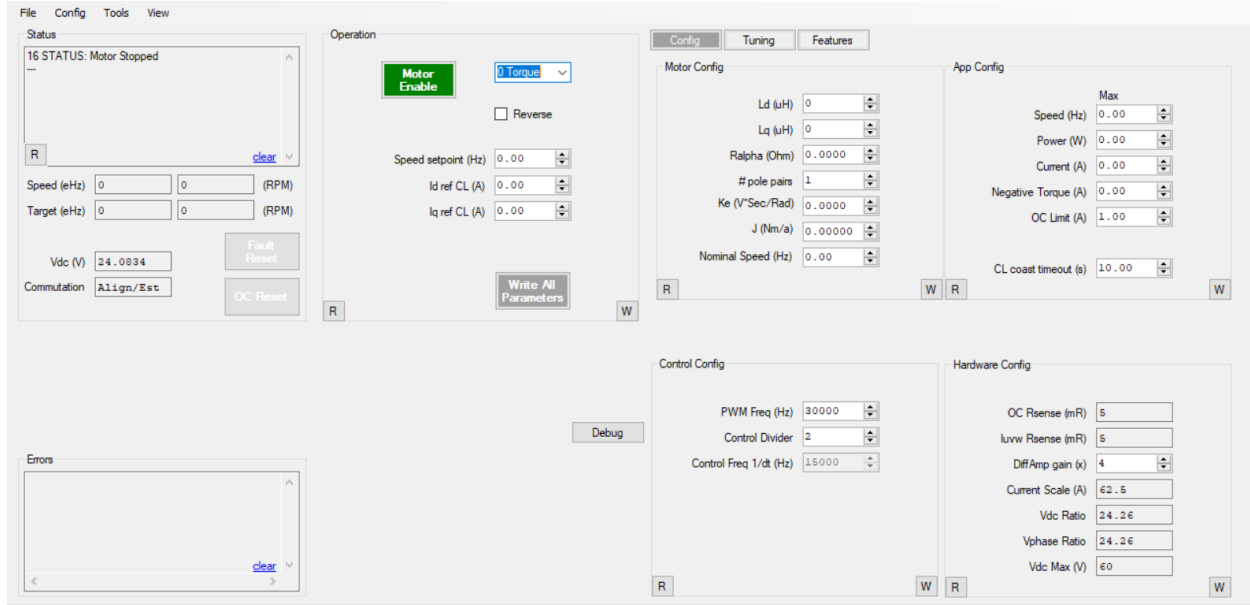
- From the IDE build the firmware image
 - If building for a PAC55xx device, make sure to install the IDE support files that can be found on active-semi.com in any PAC55xx device product folder on the software tab
 - **NOTE: If using IAR v7.x, it must be updated with new configuration files as of 2019-04-30. See .\resources\ide_support\iar for instructions.**
- Verify the build is successful with no errors before moving to the next step.
 - Some IDEs will produce some warnings which are ok to ignore.
- Make sure the Power Supply is turned on and the IDE is configured for the debugger being used
- Flash the firmware image to the device

Some IDEs won't reset the PAC device after Flashing and this has to be done manually. Turn the power supply off and back on to reset the PAC device.

Now the PAC will be running and waiting for communication from the GUI.

4 FOC GUI CONFIGURATION

Next, start the PAC FOC GUI by executing the file .resources\pac_foc_gui.exe. The application will open and be displayed as shown below.

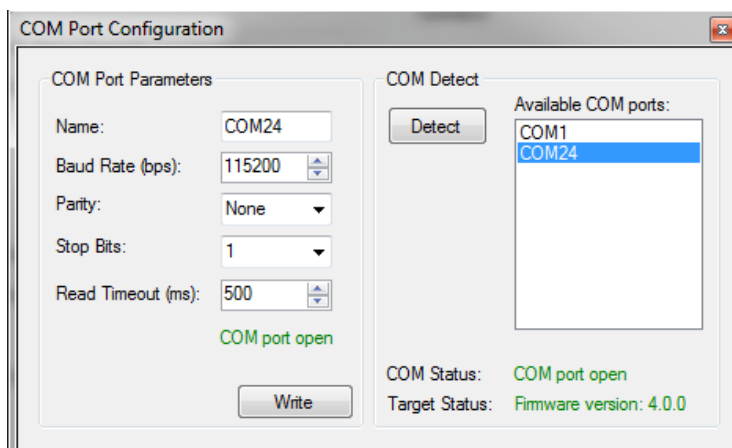


4.1 COM Port Configuration

Before the COM port is configured and the GUI connects to the PAC device, the GUI will display a “**NO COMM**” message in the bottom left corner of the GUI as shown on the right.



Configure the COM port by selecting **Config->Com Port Config...**. A dialog box will be displayed as shown below.



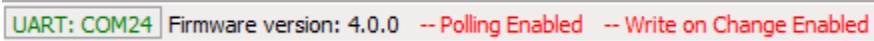
The PAC FOC FW is configured to use the default UART parameter values shown in the **COM Port Parameters** dialog box, namely 115200 bps, no Parity, and 1 Stop Bit.



Select the correct COM port from the **Available COM ports** window on the right. If the correct COM port is selected, the **COM Status** will change to read **COM port open**. Additionally, if the GUI is able to communicate with the firmware, the **Target Status** will change to display the firmware version. If the firmware version is not displayed after several seconds, reset the device by turning the power supply off and back on. Some IDEs won't reset the device after Flashing and this has to be done manually.

Once the Firmware Version is displayed, click the **Write** button to save the COM port configuration so that it will be used automatically the next time the GUI is started.

After exiting the **COM Port Parameters** dialog box, the GUI will display the COM port configuration and communication status in the bottom left corner and will automatically enable the **"Poll Status"** and **"Write on Change Enabled"** features.



UART: COM24 Firmware version: 4.0.0 -- Polling Enabled -- Write on Change Enabled

When **Poll Status** is enabled, some data, such as firmware status and motor speed, will be automatically read from the firmware periodically. When **"Write on Change Enable"** is active, new parameter values are automatically written to the firmware as soon as they are changed by the user.

4.2 Configure motor control related parameters from GUI

All the motor control related parameters can be configured from GUI. Here is a brief list of parameters to be configured from GUI (More detail please refer to "Guide_FOC_tunning.pdf" in the folder of ..\resource\guides\):

- Motor parameters (R, Ld, Lq, Ke, J and so on)
To get the above motor parameters, please refer to "Guide_Auto_Tuning.pdf" in the folder of ..\resources\guides\features
- Power board configuration including current limit, shunt resistance and so on.
- Current and speed controller setting.
- The PI gains can be automatically generated from Auto-tuning module. Please refer to "Guide_Auto_Tuning.pdf" in the folder of ..\resources\guides\features
- Add-on features for motor control.

5 OPERATING THE MOTOR

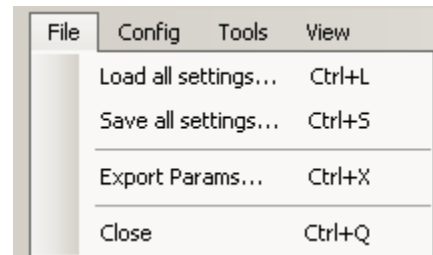
This section will describe the process to operate a motor from the GUI using a PAC FOC pre-tuned parameter file for the desired motor and PAC EVK.

If a pre-tuned parameter file has not been provided then the procedure for tuning with Qorvo “Auto-tuning” module must be followed to obtain the necessary parameters. Please refer to “Guide_Auto_Tuning.pdf” in the folder of ..resources\guides\features.

5.1 With Pre-Tuned Parameter File

5.1.1 Load The Parameter File

The FOC FW is designed so that most new motor parameters can be written from the GUI without having to recompile the firmware. If a pre-tuned parameter file is available for the motor, load it by clicking **File->Load all settings...** and navigating to the file.



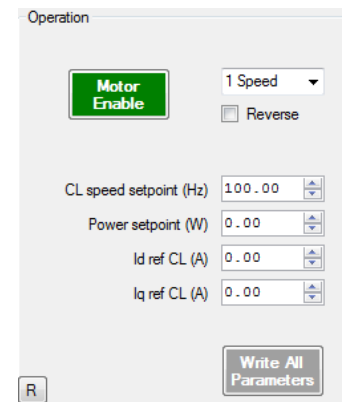
To view the parameters for the motor, click on the various Tabs in the upper right portion of the GUI such as **Config**, **Tuning**, and **Features**.

Note: Since “**Write on Change Enable**” is active, new parameter values entered from the GUI will be automatically written to the firmware as soon as they are changed by the user.

Now click “**Write All Parameters**” in the **Operation** section of the GUI as shown on the right. This will write the settings to the PAC device over the UART. The GUI will indicate the operation is in progress in the bottom left corner as shown below.

UART: COM24 Firmware version: 4.0.0 -- Polling Enabled -- Write on Change Enabled -- WRITING ALL PARAMETERS

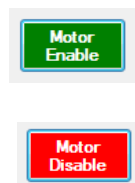
Any problems reading or writing parameters will be reported in the Errors message box and the **Write All** button will turn red. If all the parameters are written successfully, the **Write All** button will turn green.



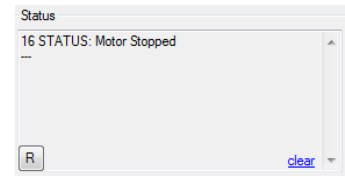
5.1.2 Running the Motor

For now, do NOT enable the motor, but just understand the mechanism to operate the motor. To enable the motor, click the green **Motor Enable** button; the button will turn Red and display **Motor Disable**. To disable the motor, click the red **Motor Disable** button.

When the motor is enabled, certain parameters will be grayed out and will no longer be accessible until the motor is disabled. This is to prevent abnormal operation and damage to the motor or PCB.



The Status section of the GUI will display various STATUS and WARNING messages. Initially, the status will show “STATUS: “Motor Stopped” as shown on the right. After enabling the motor, the STATUS will change to “STATUS: Motor Enabled”. If a warning is displayed, you can clear it by clicking on “clear”.



The GUI will also report a variety of faults in the Status message box. An Over Current (OC) fault can be cleared by clicking the **OC Reset** button. All other faults can be cleared by clicking the **Fault Reset** button.

Now, click the Motor Enable button to see the motor go through Align, Open Loop, and Closed Loop to reach the closed loop set point.

5.1.3 Next Steps

Now that the motor is operating, consult the following documents for additional information:

- Guide_FOC_Firmware_Overview.pdf – describes the functionality and architecture of the firmware
- Consult the PAC hardware User’s Guide for your specific PAC hardware.
- See other feature specific guides in .\resources\guides for SIM, Single Shunt, Stall Detect, etc.

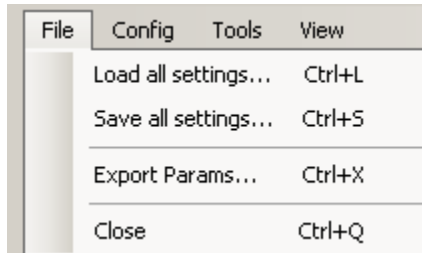
5.2 Obtaining Parameter File from Auto-tuning module

Auto-tuning feature provides the capability to identify the motor parameters together with auto-calculated PI gains. Please refer to “Guide_Auto_Tuning.pdf” in the folder of ..\resources\guides\features.

6 ADDITIONAL GUI INFORMATION

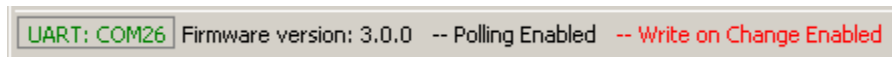
6.1 Load / Save Parameters

The GUI XML parameters files can be loaded or saved using the **Load all settings** or **Save all settings** features, respectively, accessed from the File menu.



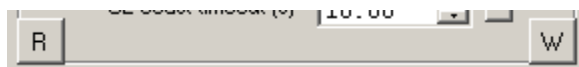
6.2 Read / Write Parameters

If the GUI is connected to the firmware, it will automatically enable the **Poll Status** and **Write on value change** features and indicate this in the bottom left corner.

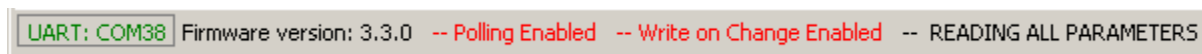


When **Poll Status** is enabled, some data, such as firmware status and motor speed, will be automatically read from the firmware periodically. When **Write on value change** is enabled, new parameter values are automatically written to the firmware immediately.

An entire parameter group can be read or written using the **R** or **W** buttons, respectively, located at the bottom of each parameter group.



All of the parameters can be read or written using the **Read all data** or **Write all data** features, respectively, accessed from the Tools menu. The GUI will indicate the operation is in progress in the bottom left corner.



Any problems reading or writing any parameter will be reported in the Errors message box and the **Write All** button will turn red. If all the parameters are written successfully, the **Write All** button will turn green.



6.3 Enabling the Motor

To enable the motor, click the green **Enable** button. When the motor is enabled, certain parameters will be grayed out and no longer be accessible until the motor is disabled. This is to prevent abnormal operation and damage to the motor or PCB. To disable the motor, click the red **Disable** button.

6.4 Resetting Faults

The GUI will report a variety of faults in the Status message box. An Over Current (OC) fault can be cleared using the **OC Reset (Ctrl+Alt+R)** feature accessed from the Tools menu or by clicking the **OC Reset** button. All other faults can be cleared using the **Fault Reset (Ctrl+Alt+F)** feature accessed from the Tools menu or by clicking the **Fault Reset** button.





CONTACT INFORMATION

For the latest specifications, additional product information, worldwide sales and distribution locations:

Web: www.qorvo.com

Tel: 1-844-890-8163

Email: customer.support@qorvo.com

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